



OPEN Evaluation of environmental impacts of cleanroom construction with a life cycle analysis approach based on energy and material consumption

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Life Cycle Assessment is an essential tool for evaluating the environmental impacts of products over their entire lifecycle. According to the United Nations Environment Programme, buildings consume the most energy throughout their lifespan. Cleanrooms, crucial in industries like biotechnology and pharmaceuticals, require significant energy, impacting ecosystems, resources, and human health. This study aims to assess the impact of enhancing wall thickness and integrating a moisture barrier layer to reduce energy usage in a cleanroom facility in Tehran dedicated to medical technology and pharmaceutical activities. The research methodology involved data collection, Building Information Modeling, Carrier HAP software for energy calculations, and SimaPro software with the Ecoinvent database for environmental impact assessment. Results showed that modifying building materials and energy use can decrease contributions to the global warming index by 1 to 6%. Using 20-cm-thick walls with cement mortar increased global warming index pollution from 29 to 57.82% post-material adjustments. Increasing concrete wall thickness and adding insulation can reduce energy consumption and environmental impacts over a building's 50-year lifespan, resulting in a 4.5% reduction across all environmental indices. This research emphasizes the importance of long-term sustainability in construction projects and highlights the benefits of thicker walls in mitigating environmental impacts.

Keywords Environmental impact assessment, Cleanroom, Energy, Sustainability

In recent decades, the ecological impact of buildings has gained significant attention from researchers, policymakers, and organizations due to concerns about environmental degradation and resource depletion¹. This has resulted in a growing demand for precise assessments of environmental impacts using methods like Life Cycle Assessment (LCA) and Environmental Impact Assessment (EIA)². These methodologies are increasingly utilized in industry and policy to evaluate effects on ecosystems, natural resources, and human health. Life Cycle Assessment assesses a product's environmental impact throughout its lifespan, from resource extraction to emissions, while Environmental Impact Assessment systematically identifies and evaluates potential environmental effects of proposed projects³. Both methods follow international standards ISO 14040 and ISO 14044, offering structured frameworks for informed decision-making⁴. Buildings contribute to adverse environmental impacts through significant resource and energy consumption during operational lifetimes. The construction sector prioritizes sustainable technologies and practices in response to rising environmental concerns. Cleanrooms adhere to strict regulations and are particularly energy-intensive due to their HVAC systems, often consuming more energy per square foot than standard commercial buildings⁴.

The importance of cleanrooms in semiconductor manufacturing is highlighted by Wainer et al. (2018), who emphasize their role in reducing particle contamination through advanced filtration systems like HEPA and ULPA. Their study underscores that effective design, construction, and maintenance are crucial for cleanroom performance, though predicting their behavior under different operational conditions remains challenging. This

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research introduces new tools for estimating contamination levels, which is vital for maintaining cleanliness standards essential for high-quality semiconductor production⁵. Similarly, William et al. (2019) investigated energy efficiency in hospital buildings in Egypt, revealing a troubling increase in energy consumption in healthcare facilities and evaluating energy-saving measures. Through Design Builder simulations of a hospital in Alexandria, they identified strategies such as a Dedicated Outdoor Air System (DOAS) that can significantly lower HVAC Energy usage, highlighting the importance of sustainable practices in building design relevant to cleanroom construction where energy-intensive HVAC systems are prevalent⁶. Additionally, Jiaowu et al. (2020) performed a performance analysis of filtration and air conditioning processes in semiconductor cleanrooms, noting the high energy demands for maintaining optimal air change rates and airflow resistance. Their findings indicate that high particulate concentrations worsen the load on filtration systems, increasing energy consumption. They propose an innovative air filtration and conditioning system that filters outdoors and returns air separately, achieving 25.8% to 45.0% energy savings due to reduced airflow resistance. The study also recommends optimal return airflow rates to prevent over-pressurization, potentially saving 32.9% to 50.7% compared to conventional systems⁷.

In 2022, Rezvan et al. conducted a significant study titled "Optimizing HVAC Designs in Indoor Environments: Advancing Energy and Environmental Benefits of Cleanrooms," focusing on the need to reduce energy consumption in cleanroom construction to meet carbon reduction targets set by the Paris Agreement. Cleanrooms require strict control over temperature, humidity, and CO₂ emissions, making this research particularly relevant. The study explored three control configurations (Options A, B, and C) and found Option B to be the most effective, achieving a 25% reduction in energy consumption and CO₂ emissions while maintaining setpoint temperatures and acceptable humidity levels. This highlights the potential for optimized HVAC systems to improve operational efficiency and reduce the environmental footprint of cleanroom facilities⁸. In mid-2022, Al-Ghamdi et al. examined the "Impact of Architectural Design Parameters on Thermal Comfort and Energy Consumption in Higher Education Buildings." Their research identified key architectural design parameters (ABDPs) that enhance thermal comfort and lower energy use. Adjustments such as cooling setpoint temperature and ceiling construction could decrease operational temperatures by 14.2 and 20.0%, respectively. These insights suggest that incorporating specific architectural considerations into cleanroom designs could promote sustainability and support the evaluation of environmental impacts through life cycle analysis⁹. Salim et al. (2023) provided an overview in their study "Life Cycle Assessment of Building Materials: Methods, Applications, and Future Directions for Sustainable Decision Making." This research delves into Life Cycle Assessment methodologies relevant to building materials, which are essential for understanding the environmental impacts of cleanroom construction. It emphasizes the importance of standardizing data quality in decision-making processes. It encourages collaboration between industry and research sectors to foster sustainable construction practices, particularly regarding material selection and energy consumption in cleanroom environments¹⁰. In early 2024, Mazur et al. published a study titled "Assessing External Wall Systems in the Passive House Standard: Life Cycle Assessment Analysis of Materials, Thermal Properties, and Environmental Impact," which offers valuable insights into sustainable construction practices relevant to cleanroom design. The research evaluates various external wall systems in single-family homes that meet the Passive House standard, finding that timber structures significantly outperform traditional masonry walls in energy consumption, CO₂ emissions, and water usage. This suggests a shift towards more sustainable materials in construction, with implications for cleanroom environments where similar strategies could lead to substantial reductions in energy and resource consumption, aligning to evaluate environmental impacts through life cycle analysis¹¹. Similarly, Zuo et al. conducted a study in 2024 titled "Impact of Pollutant Emissions on Fresh Air Intake in Cleanroom Workshops." This research uses Computational Fluid Dynamics (CFD) methods to examine how exhaust pipe height, velocity, barrier structure, and wind direction influence pollutant emissions in cleanroom settings. The findings indicate that optimizing these parameters can significantly reduce pollutant concentrations in fresh air intakes, improving indoor air quality. This study highlights the importance of adequate ventilation strategies in cleanrooms, emphasizing that better pollutant control enhances environmental conditions and protects worker health. By incorporating these insights into our life cycle analysis approach, we can more accurately assess how ventilation design choices affect energy consumption and material usage in cleanroom construction¹².

Cleanrooms are essential for maintaining optimal temperature, humidity, and air filtration conditions. According to ISO standard 14644-1, clean rooms are designed to control airborne particle concentrations, minimizing particle introduction, generation, and retention¹³. These environments are critical in industries such as medical and pharmaceuticals, where they ensure the safety, efficacy, and quality of biotechnological vaccines. However, the construction of clean rooms relies heavily on materials like cement and blocks, which have significant environmental impacts¹⁴. The construction industry is among the largest consumers of natural resources and producers of waste globally¹⁵. Extracting raw materials for cement production, including limestone, sand, and clay, can lead to habitat destruction, pollution, and high energy consumption. Additionally, the production and transportation of these materials contribute significantly to greenhouse gas emissions^{16,17}. Clean rooms, particularly in pharmaceutical facilities, have notably high energy demands, consuming up to 25 times more energy for heating and cooling than standard commercial buildings. In response to these challenges, ISO 14644-16 recommends enhancing energy efficiency throughout the cleanroom lifecycle¹⁹. Energy Building Modeling (BEM) techniques leverage BIM software to analyze and optimize energy performance across building components. These methods are crucial for reducing energy consumption through comprehensive evaluations of building energy utilization²⁰. Numerous studies have explored how building materials affect the environment in various building contexts. Strategies to enhance energy efficiency and reduce consumption have been proposed for cleanrooms and traditional buildings. However, there still needs to be a gap in conducting concurrent assessments that integrate life cycle evaluations of materials with analyses of environmental impacts from thermal and cooling energies, particularly within cleanroom environments. This study introduces an

innovative method to thoroughly examine how energy use and material consumption affect the environment in cleanrooms. To achieve its goals, it utilizes BIM and Simapro software. BIM allows for creating accurate virtual models of cleanroom environments, enabling the simulation of various scenarios to evaluate energy efficiency. Simapro, a sophisticated tool for life cycle assessment, offers an in-depth analysis of energy, materials, and construction procedures. Ultimately, these tools aim to guide the construction industry towards increased sustainability and flexibility, ensuring environmental preservation for future generations.

Research methodology

This study focuses on a cleanroom structure to assess the impact of material modifications and wall insulation on energy consumption and environmental footprint. The chosen materials' comprehensive life cycle assessment evaluates their sustainability and long-term performance. Essential data for energy calculations encompass weather conditions, geographical information, and room specifics, including occupancy patterns, work intensity, and lighting power. This holistic approach ensures that all relevant factors influencing energy consumption are considered. Energy analysis and evaluation are conducted using advanced software tools such as Carrier HAP (Hourly Analysis Program) and AECOSim. These tools facilitate detailed simulations of heating and cooling loads based on the input parameters, allowing for an accurate representation of energy use under varying conditions. Revit is essential in BIM, providing precise and detailed modeling of building components. The models created in Revit offer comprehensive information on various aspects of the building within an interactive and integrated environment, enhancing collaboration among stakeholders. A significant application of Revit is facilitating building information exchange, mainly through its support of the gbXML file format, which is recognized as the standard for green building information exchange. Additionally, the Industry Foundation Classes (IFC) format is a robust standard for exchanging building information, providing a detailed model of building geometry. Integrating IFC with gbXML enables a comprehensive view of a building's energy performance by combining data on geometry, materials, mechanical systems, and climatic conditions into a single model. This integration allows for more precise energy consumption analyses and helps identify additional energy-saving opportunities that may arise from design modifications or operational adjustments. The study sets a 50-year timeframe for evaluating the life cycle assessment of materials and energy consumption, aligning with the expected lifespan of the case study building. This long-term perspective is critical for understanding how initial material choices and design decisions impact energy efficiency over time. Focusing mainly on heating energy provided by boilers and cooling energy provided by chillers within the cleanroom environment is crucial for understanding overall energy dynamics. These devices consume different quantities of gas and electricity, highlighting the importance of monitoring their operational efficiency. The analysis includes a detailed breakdown of energy consumption rates for heating and cooling systems, providing insights into how different configurations and operational strategies can influence total energy use.

The Carrier software outputs data on electricity and gas consumption rates, crucially integrated with equipment specifications and client-provided catalogs. This integration is vital input for the software analysis, enabling a more nuanced understanding of how equipment performance correlates with overall energy consumption in the cleanroom environment. By analyzing these consumption patterns, we can identify potential areas for optimization, ultimately contributing to enhanced energy efficiency and reduced environmental impact.

This study's main objective is to investigate the effect of cleanroom wall thickness on energy consumption and environmental impacts resulting from design changes. Cleanrooms require specific designs due to the need to closely control ecological conditions, including temperature and contamination, which can lead to high energy consumption. In this study, two scenarios for designing cleanroom walls with a useful life of 50 years are examined.

1. Scenario 1: In this scenario, the structural design features walls with a thickness of 10 cm, utilizing blocks that measure 10 cm in width.
2. Scenario 2: Increasing the wall thickness to 20 cm significantly enhances the design parameters. This modification employs blocks measuring 20 cm in width and incorporates an insulating layer composed of bitumen.

The SimaPro software conducts life cycle assessments of materials and analyzes the environmental impacts of heating and cooling energies in scenarios one and two. These assessments conform to international standards ISO 14040 and ISO 14044, utilizing the Ecoinvent database within the software for specialized data compilation and evaluation. SimaPro software employs the ReCiPe 2016 Endpoint (H) method to assess environmental impacts associated with building materials, heating, and cooling energies. This method predicts potential environmental effects from these factors and evaluates their importance. Recommendations are then proposed to mitigate these adverse impacts. Using characterization and weighting techniques, the ReCiPe 2016 Endpoint (H) method quantifies the implications of studies on ecosystems, human health, and resources. The evaluation and comparison of the environmental effects of heating and cooling energies in cleanrooms, scenarios one and two, are illustrated in the research flowchart presented in Fig. 1. This research uses standard protocols such as ISO 14044-ISO 14044 as a guide and main framework to create a systematic structure and guide the assessment process. Next, the objectives and scope of the research are also specified, including determining the study's overall and specific objectives, and the system boundary is clearly defined to identify the processes and activities related to the research topic. In addition, the functional unit that is the basis for measuring environmental impacts is determined.

Goal and scope definition

The Life Cycle Assessment system boundary for the cleanroom building is comprehensively defined to encompass all stages, from extraction and processing of raw materials to construction to demolition and disposal, over an

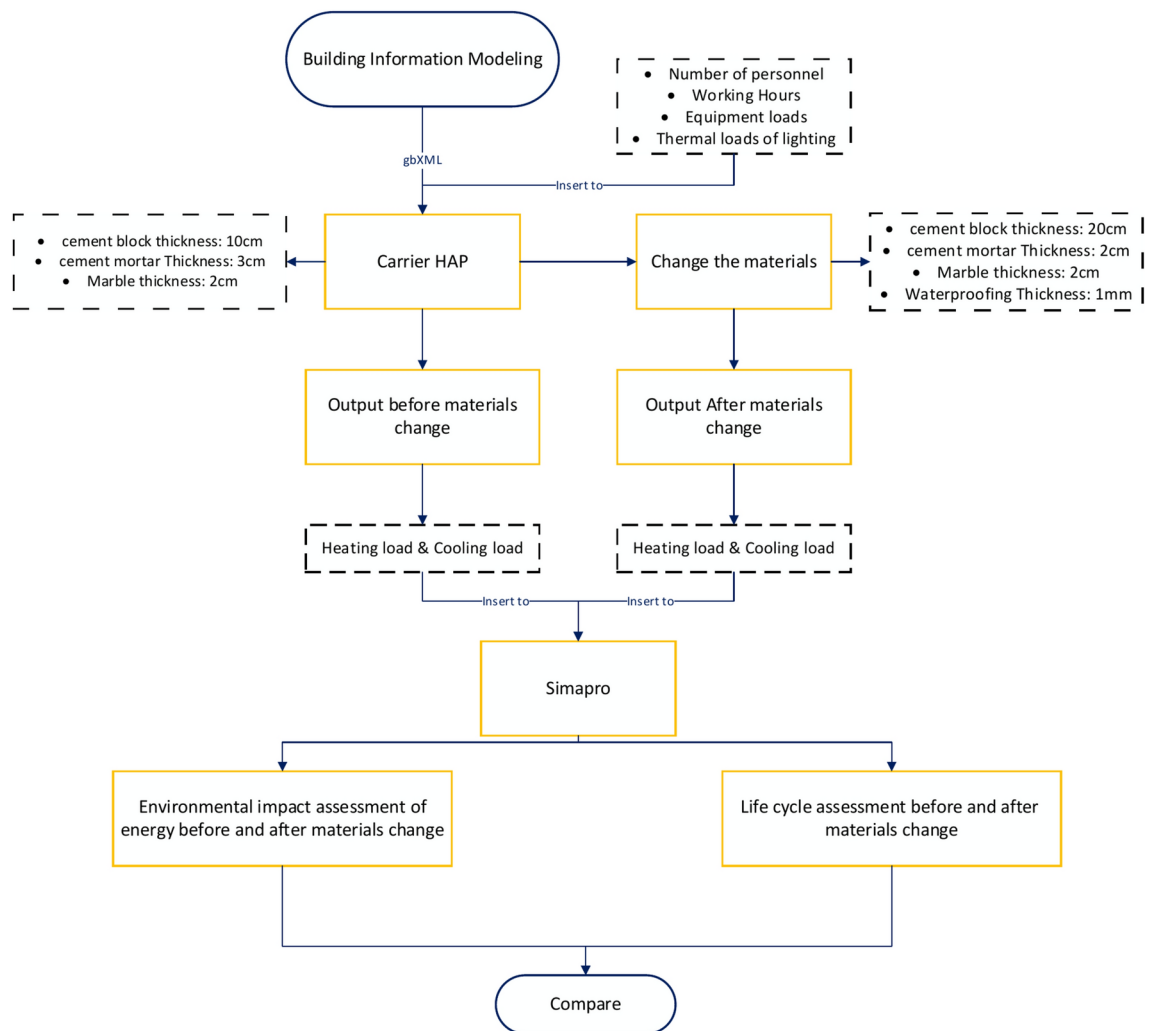


Fig. 1. Research methodology steps.

expected lifetime of 50 years. This life cycle assessment is performed at SimaPro using the ReCiPe 2016 Endpoint (H) method. This holistic approach thoroughly evaluates the environmental impacts of each cleanroom's life cycle phase¹⁸. The assessment begins with extracting raw materials, including the environmental implications of mining, harvesting, or otherwise procuring the necessary resources. This stage is critical as it sets the foundation for the entire life cycle, influencing the availability of materials and the ecological footprint associated with their acquisition. Following raw material extraction, the processing phase is examined. This involves analyzing the energy consumption, emissions, and waste produced while manufacturing building materials. The production processes for concrete, steel, glass, and insulation materials are scrutinized to quantify their environmental impacts. The transportation of these materials to the construction site is also included in this boundary, as it contributes significantly to greenhouse gas emissions and resource depletion¹⁹. The construction phase is another vital Life Cycle Assessment component. This section evaluates not only the direct environmental impacts of construction activities such as land disturbance, resource use, and waste generation but also considers the efficiency of construction practices. The installation of various building components, including walls, HVAC systems, and cleanroom technologies, is analyzed to understand how these choices influence overall sustainability. Once the cleanroom is operational, the assessment focuses on its ongoing operation and maintenance. This includes evaluating energy consumption for heating, cooling, ventilation, and lighting over the lifespan. The analysis also incorporates maintenance activities that may require additional resources and generate waste. Energy savings from enhanced wall thickness and insulation are particularly emphasized, as they are crucial in reducing operational costs and minimizing environmental impact. The study addresses demolition and disposal scenarios at the end of the cleanroom's life cycle. This phase assesses how materials are deconstructed, recycled, or sent to landfills, thus evaluating potential waste generation and opportunities for material recovery²⁰. The implications of heightened material use throughout the cleanroom's life cycle are critically analyzed to ensure that all aspects of sustainability are considered. Figure 2 shows the relationship between midpoints and endpoints in the 2016 recipe method.

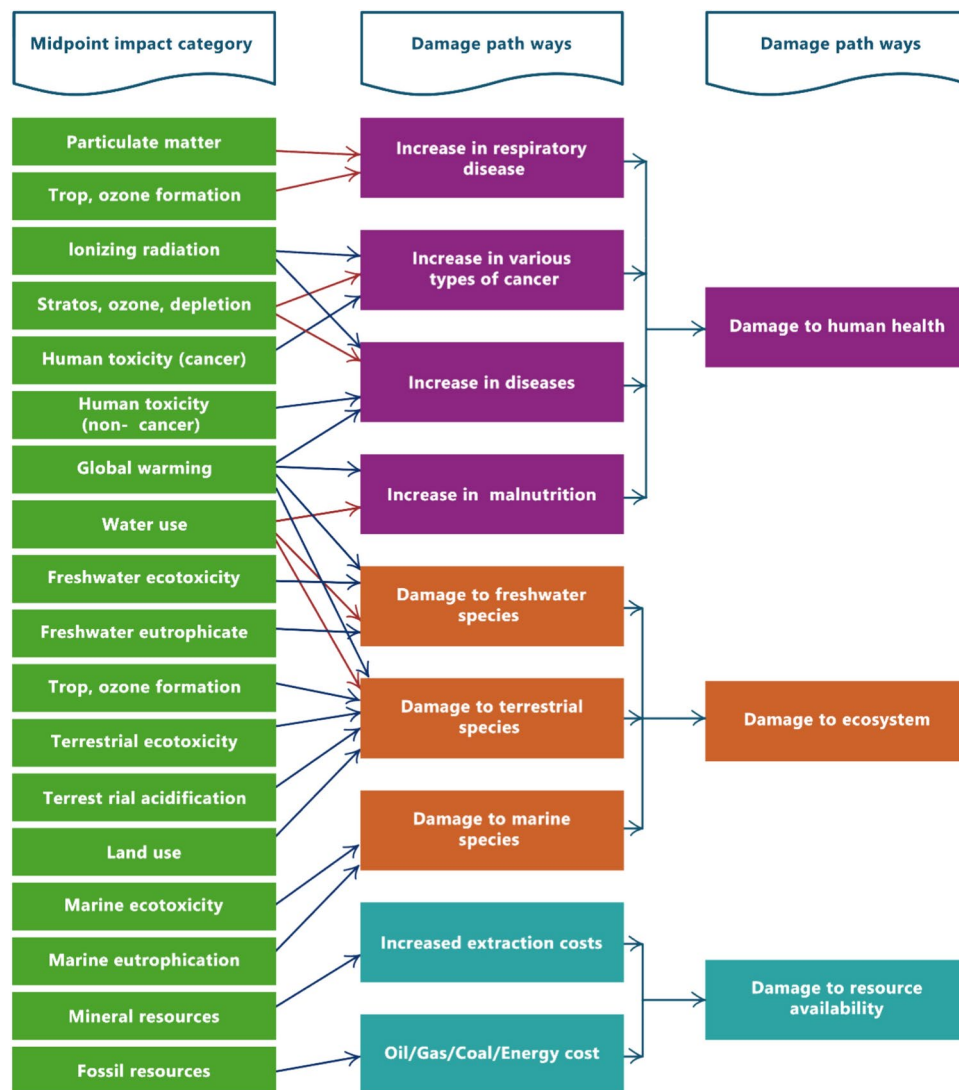


Fig. 2. Relationship between midpoints and endpoints in the recipe 2016 method2-2 system boundary.

This study comprehensively defines the system boundary for the Life Cycle Assessment of a cleanroom in Tehran dedicated to medical technology and pharmaceutical activities. It includes all raw materials for final destruction and disposal over the 50-year life of the building. This consists of the environmental impacts resulting from the production and processing of raw materials, construction operations, energy consumption and maintenance throughout the life cycle, and finally, destruction and consumption of materials. This study examines the impact of design choices on environmental sustainability and operational efficiency for cleanrooms and aims to determine how these choices affect energy consumption and production. The garden is at each stage. This Life Cycle Assessment study employs a cradle-to-grave approach, illustrated in Fig. 3.

Functional unit

The choice of performance unit in life cycle assessment is critical because it provides a basis for comparing and determining the environmental impacts associated with the system under consideration. In this study, the performance unit is defined according to the guidelines of ISO 14044. This standard emphasizes precisely defining the performance unit to ensure consistency and comparability in environmental assessments. The functional unit in this project is for a clean room building with an area of 2,400m².

Case study

The case study examines the design of a cleanroom within a pharmaceutical facility. The building covers a total area of 2,400 square meters, with the cleanroom occupying 941 square meters and a height of 24 m. It includes various rooms and is designed to have a lifespan of 50 years. The building is located at 35.4 degrees latitude and 51.5 degrees longitude, 5,710 feet above sea level. The design temperatures are 104°F in summer and 18°F in winter. This facility, dedicated to medical technology and pharmaceutical operations, is located in

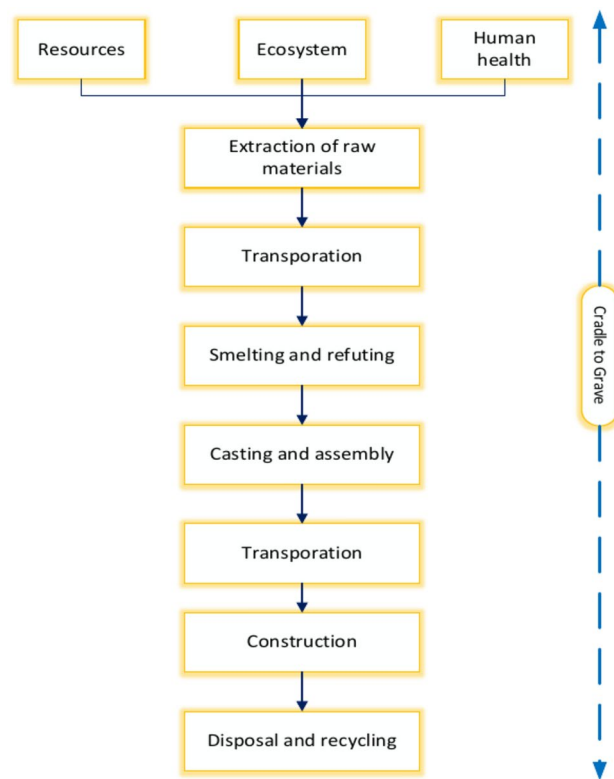


Fig. 3. Life cycle assessment system boundary.

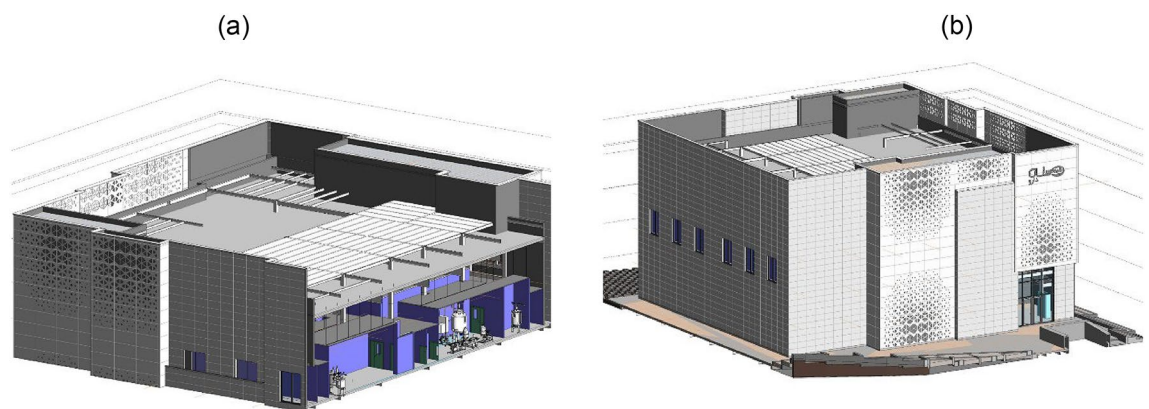


Fig. 4. (a) Autodesk 3D north section view; (b) Autodesk 3D north view.

Tehran's Pardis Technology Park. Figure 4a shows the Autodesk 3D north section view, while Fig. 4b presents the Autodesk 3D north view.

The cleanroom building plan is segmented into seven distinct zones, each reflecting unique characteristics identified during the design phase. The division into zones is influenced by room classification, air handler placement, room size, and differences in humidity and temperature. Figure 5a illustrates these seven zones.

Room temperatures are tailored to operational requirements. In our case study, most areas are set at 24 degrees Celsius with 40% relative humidity, except for the cold room. This temperature choice aligns with client preferences and operational needs. Specifically, the cleanroom's ampoule-filling operation necessitates a constant temperature of 24 degrees Celsius. Figure 5b provides a visual representation of the temperature distribution.

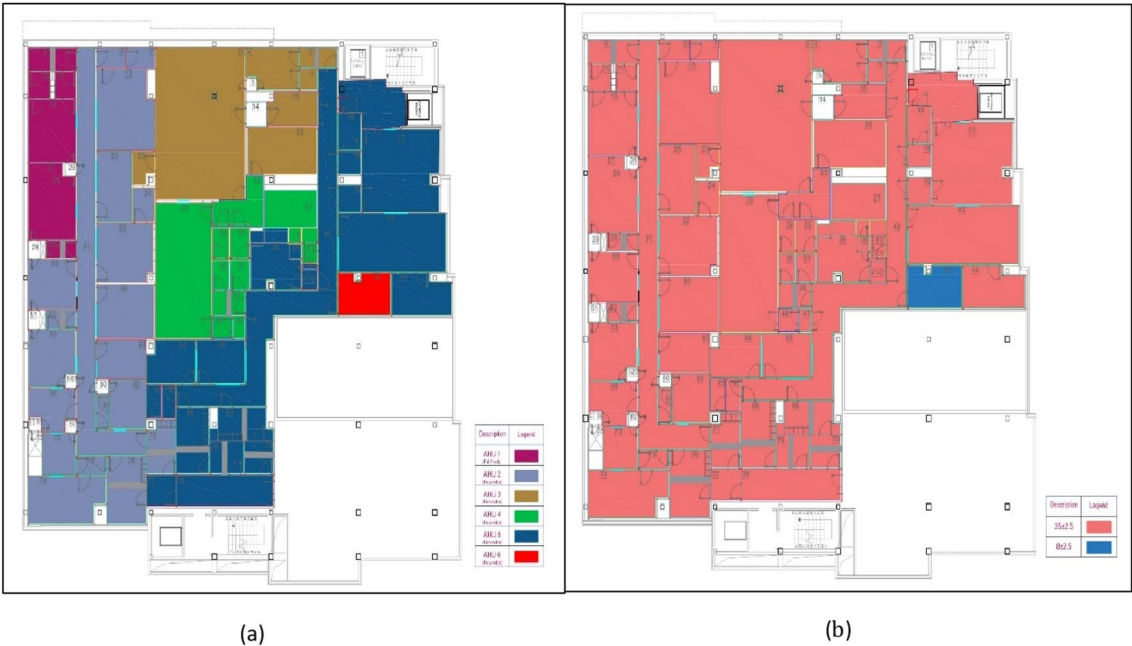


Fig. 5. (a) Zone plan; (b) Temperature plan.

No	The outlet temperature from the air conditioner (F)	Heating load (MBH)	Airflow (CFM)
Zone 1	70.8	7.4	1349
Zone 2	67	155.6	9191
Zone 3	66.6	8.6	4909
Zone 4	66.3	4.4	3143
Zone 5	67.3	5.5	5586
Lobby	70.3	29.5	10,884

Table 1. Heating load analysis before materials change (scenario 1).

No	The outlet temperature from the air conditioner (F)	Cooling load (MBH)	Airflow (CFM)
Zone 1	50.5	21	1349
Zone 2	50.3	165.8	9191
Zone 3	52.5	87.8	4909
Zone 4	53.1	56.7	3143
Zone 5	53	105.8	5586
Lobby	50.2	195.3	10,884
Cold room	40	8	2000

Table 2. Cooling load analysis before materials change (scenario 1).

Discussion
Inventory analysis

Analysis of heating and cooling energy before material changes (Scenario1)
Meteorological and climatic data relevant to the building’s location are obtained from the meteorological organization. Based on the highest recorded summer temperature, the summer design temperature is 104 degrees Fahrenheit. In comparison, the winter design temperature is established at 18 degrees Fahrenheit, derived from the lowest recorded winter temperature. Each room is maintained at a consistent temperature of 55 degrees Fahrenheit. Tables 1 and 2 provide detailed information on heating and cooling energy, including airflow rates for each zone before material modifications (Scenario 1). Tables 3 and 4 display the respective data following the material changes.

No	The outlet temperature from the air conditioner (F)	Heating load (MBH)	Airflow (CFM)
Zone 1	70.8	6.90	990
Zone 2	67	149.9	8810
Zone 3	66.6	8.2	4790
Zone 4	66.3	4.2	3015
Zone 5	67.3	5.3	5490
Lobby	70.3	27.1	9050

Table 3. Heating load analysis after materials change (scenario 2).

No	The outlet temperature from the air conditioner (F)	Cooling load (MBH)	Airflow (CFM)
Zone 1	50.5	19	990
Zone 2	50.3	158.6	8810
Zone 3	52.5	85.1	4790
Zone 4	53.1	54.2	3015
Zone 5	53	101.1	5490
Lobby	50.2	183.2	9050
Cold room	40	8	2000

Table 4. Cooling load analysis after materials change (Scenario 2).

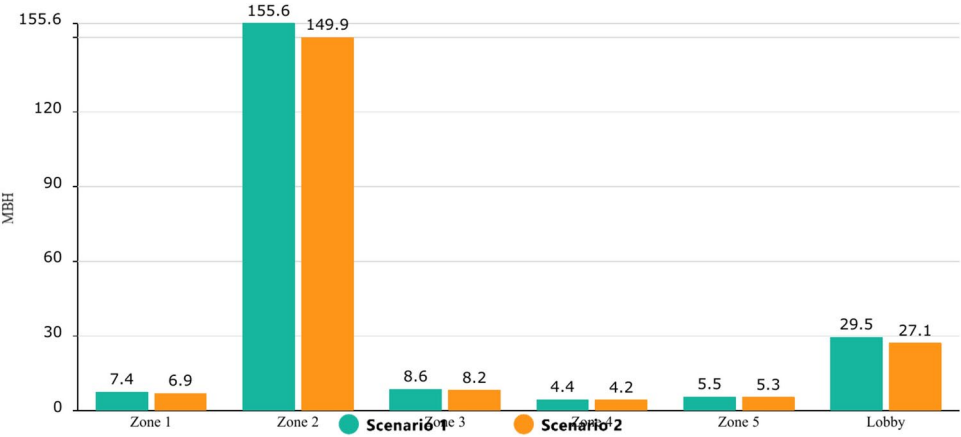


Fig. 6. Comparison of heating load in scenario1 and scenario 2.

Energy analysis for heating and cooling after material changes (Scenario 2)

Cleanrooms accommodate varying equipment heights, resulting in zones with different ceiling heights. Rooms with taller ceilings undergo more frequent air exchanges, which increases energy usage. Conversely, zones with lower ceilings demand less energy. The lobby shows a significant energy contrast before (Scenario 1) and after material changes (Scenario 2), as it lacks prefabricated panels and interior walls in cleanrooms, exposing it to temperature fluctuations. Unlike cleanrooms, which have multiple-layered walls and panels, they are less affected by external thermal changes directly impacting temperature regulation.

Figures 6, 7 illustrate the comparison of heating and cooling energy consumption in the cleanroom before (Scenario 1) and after material changes (Scenario 2). The vertical axis shows the heating load value, and the horizontal axis shows each zone. The data clearly show a reduction in energy usage for heating and cooling following the adjustments in materials.

Energy consumption of electricity and gas in Scenario 1 and Scenario 2

Table 5 displays the combined electricity and gas usage before (Scenario 1) and after material alterations (Scenario 2) across each zone during the building’s 50-year operational span, factoring in the average efficiency of chillers and boilers.

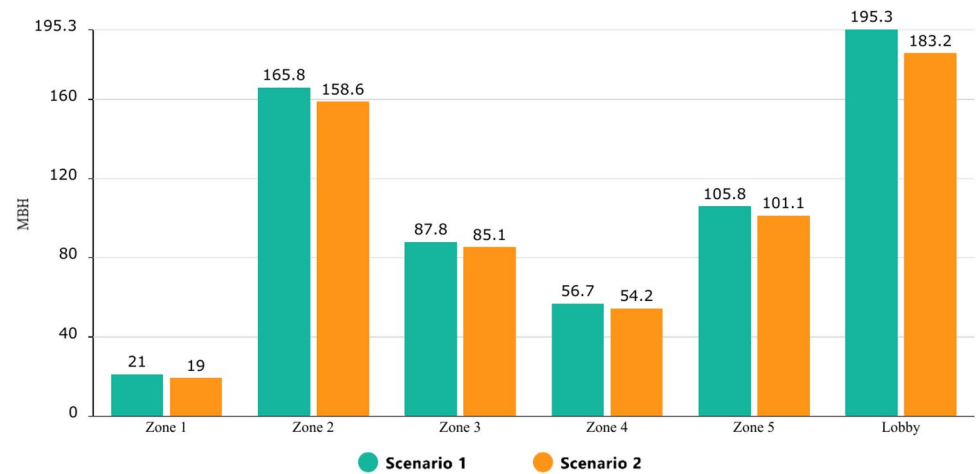


Fig. 7. Comparison of cooling load in scenario1 and scenario 2.

No	Electricity consumption in Scenario 1	Electricity consumption in Scenario 2	Gas consumption in Scenario 1	Gas consumption in Scenario2
Zone 1	1,029,000	931,000	5,439,000	5,071,500
Zone 2	8,124,200	7,770,000	114,366,000	110,176,500
Zone 3	4,302,200	4,169,760	6,320,160	6,027,000
Zone 4	2,778,300	2,655,800	3,234,000	3,087,000
Zone 5	5,184,200	4,953,900	4,042,080	3,895,500
Lobby	9,569,700	8,976,800	21,756,000	19,918,500
Cold Room	996,800	996,800	–	–

Table 5. Electricity and Gas Consumption in Each Zone in Scenario 1 and Scenario 2.

No	Materials	Unit	Total weight of materials
1	Cement mortar for the 10 cm block (Scenario 1)	M³	97
2	Cement mortar for the 20 cm block (Scenario 2)	M³	193
3	Block 10 cm (Scenario 1)	Kg	39,263
4	Block 20 cm (Scenario 2)	Kg	78,252
5	Insulation (Scenario 2)	Kg	150

Table 6. Material and waste inventory for construction scenarios of the cleanroom project.

Waste management information

In the life cycle assessment of the cleanroom construction project, two scenarios were evaluated, with careful consideration of the waste generated in each. These waste quantities were not recorded as separate activities in the SimaPro software but were instead incorporated into the life cycle inventory (LCI) calculations for each material. Table 6 outlines the type and quantity of waste generated at each project stage, providing a detailed breakdown for both scenarios. Given the importance of waste management in environmental impact assessments, these quantities were thoroughly accounted for in the analysis. This study employs landfilling as the primary method for managing construction waste. As a widely used and traditional approach, particularly for construction materials, landfilling reduces waste volume and supports public health. The process involves depositing waste in designated sites to minimize environmental pollution and health risks.

Materials change information

The planned cleanroom building features an increase in block thickness from 10 to 20 cm. This enhancement involves incorporating insulation to lower energy consumption despite the resulting rise in material usage. Table 7 provides a breakdown of the materials utilized.

Environmental impact assessment

The primary life cycle assessment software methods are characterization, damage assessment, normalization, weighting, and final scoring. Characterization evaluates the environmental impact of each emitted substance and categorizes them into specific impact classes, a critical step for identifying various environmental impact

No	Materials	Unit	Specific weight	Quantity	Total weight of materials
1	Cement mortar for the 10 cm block (Scenario 1)	M ³	2100	—	97
2	Cement mortar for the 20 cm block (Scenario 2)	M ³	2100	—	193
3	Block 10 cm (Scenario 1)	Kg	4.5	8725	39,263
4	Block 20 cm (Scenario 2)	Kg	9	8725	78,252
5	Insulation (Scenario 2)	Kg	10	15	150

Table 7. Materials change information.

No	Indicators	No	Indicators
1	Global warming, Human health	12	Terrestrial ecotoxicity
2	Global warming, Terrestrial ecosystems	13	Freshwater ecotoxicity
3	Global warming, Freshwater ecosystems	14	Marine ecotoxicity
4	Stratospheric ozone depletion	15	Human carcinogenic toxicity
5	Ionizing radiation	16	Human non-carcinogenic toxicity
6	Ozone formation, Human health	17	Land use
7	Fine particulate matter formation	18	Mineral resource scarcity
8	Ozone formation, Terrestrial ecosystems	19	Fossil resource scarcity
9	Terrestrial acidification	20	Water consumption, Human health
10	Freshwater eutrophication	21	Water consumption, Terrestrial ecosystem
11	Marine eutrophication	22	Water consumption, Aquatic ecosystems

Table 8. Classes of effect in simapro ReCiPe 2016 endpoint (H) method.

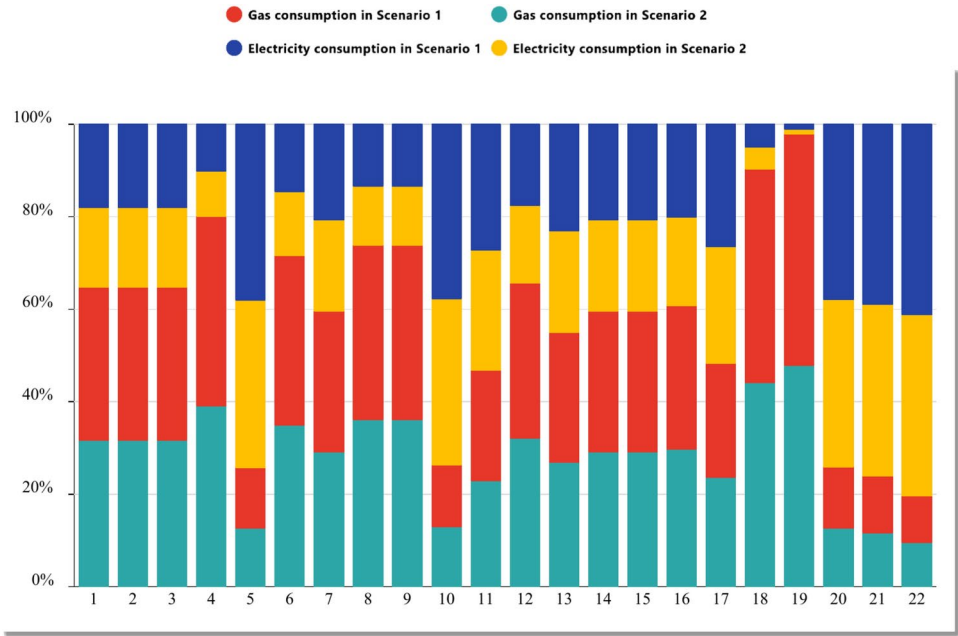


Fig. 8. Environmental impacts assessment of energy consumption based on various indicators.

sources related to the project. The Life Cycle Assessment software includes 22 impact indicators, as shown in Table 8. Following characterization and normalization, weighting assesses the relative importance of each impact category in three key areas: human health, ecosystems, and resources. This step assigns weight coefficients to express the relative significance of different impact categories²¹.

Life cycle impact assessment

Characterization

Figure 8 illustrates that gas consumption has a more significant environmental impact than electricity regarding global warming potential. In Fig. 8, the vertical axis shows the percentage of destruction, and the horizontal axis

shows each indicator. Before the material changes (Scenario 1), gas accounted for 33%, and electricity accounted for 18.06% of the total global warming impact. These figures drop by approximately 4% after the material modifications (Scenario 2). The significant effect of gas is mainly due to methane, a powerful greenhouse gas. Methane combustion releases it into the atmosphere, worsening global warming. Additionally, methane leaks during natural gas extraction, transportation, and storage further elevate atmospheric methane levels. In many areas, the carbon intensity of natural gas is higher than that of electricity, especially when electricity comes from renewable sources or low-carbon technologies like hydropower. Therefore, relying on natural gas increases greenhouse gas emissions compared to cleaner energy alternatives²².

Pollution from gas consumption decreases by 5%, and from electricity, it decreases by 1%. Burning natural gas, in particular, releases pollutants like nitrogen oxides and volatile organic compounds, contributing to ozone layer depletion. Before material changes (Scenario 1), electricity contributes 38.02% and gas 13.17% to pollution in the ionizing radiation indicator. Following the changes, pollution from electricity decreases by 5% and from gas by 1%. Recent advancements in electricity generation, such as enhanced power plant efficiency, emission control technologies, and integration of renewable energy sources, significantly mitigate ionizing radiation pollution. Electricity is more effective in reducing this pollution than gas due to emission intensity, technological advancements, regulatory measures, and the energy mix²³.

Before the material changes (Scenario 1), ozone formation indicators related to human health, terrestrial ecosystems, and soil acidification showed that gas contributed 37%, while electricity contributed less than 14%. Following the adjustments in gas materials, there was a decrease of approximately 4.5% in gas-related pollution and a reduction of about 3% in pollution from electricity. Generally, electricity consumption leads to lower levels of ozone formation-related pollution and its associated impacts on human health compared to gas consumption. Gas emissions, including nitrogen oxides (NO_x) and volatile organic compounds (VOCs), are significant precursors to ground-level ozone formation. Conversely, electricity generation from cleaner sources like renewable energy or nuclear power emits fewer pollutants that contribute to ozone formation. Ground-level ozone poses significant risks to plant growth and ecosystem health. Electricity produced from gas-fired power plants releases pollutants that contribute to ozone formation. To mitigate these impacts on terrestrial ecosystems, decreasing pollution from gas consumption and increasing reliance on cleaner electricity sources is essential. Gas combustion releases sulfur dioxide (SO₂) and nitrogen oxides, leading to soil acidification and acid rain deposition. In contrast, electricity generated from cleaner sources emits lower levels of these pollutants, thereby helping to reduce soil acidification in terrestrial ecosystems. Emphasizing the reduction of pollution from electricity consumption through cleaner energy sources or emission control technologies can effectively minimize the adverse effects of soil acidification compared to exclusively targeting reductions in gas consumption²³.

Gas consumption contributes 12.88 and 22.85% to freshwater and marine eutrophication indicators, respectively. It generally emits fewer nitrogen and phosphorus compounds than electricity production from specific sources. Despite producing nitrogen oxides through combustion, which accumulate in aquatic ecosystems and contribute to eutrophication, gas consumption typically has a lesser overall impact on nutrient pollution in these environments than specific electricity generation methods. Electricity consumption significantly reduces pollution in various environmental and health toxicity indicators compared to gas consumption. Specifically, it contributes 13.5, 5, 10, 10, and 10% less pollution in terrestrial, freshwater, marine, carcinogenicity, and non-carcinogenic toxicity, respectively. Gas consumption primarily involves burning natural gas, which emits pollutants such as nitrogen oxides, sulfur dioxide, and particulate matter into the atmosphere. These pollutants contribute to ecotoxicity in terrestrial, freshwater, and marine ecosystems. Moreover, natural gas combustion releases CO₂, exacerbating climate change. Electricity generation emits CO₂, yet its emissions are generally lower than gas emissions, particularly when utilizing renewable energy sources. Gas consumption releases toxic chemicals and volatile organic compounds, posing risks to human health and the environment regarding carcinogenic and non-carcinogenic effects. Cleaner electricity production mitigates these emissions. On the other hand, extracting, transporting, and burning natural gas can adversely affect the environment through habitat destruction, water contamination, and air pollution. Overall, electricity generated from cleaner sources tends to have a reduced environmental footprint throughout its lifecycle²⁴.

Regarding resource scarcity and the depletion of fossil resources, electricity consumption causes minimal environmental impact compared to gas, contributing only 5% and 1% to pollution, respectively. Extracting natural gas can lead to environmental degradation, habitat destruction, and depletion of fossil resources. On the other hand, electricity can be generated from diverse renewable sources such as solar, wind, hydroelectric, and geothermal energy, which generally require fewer mineral resources than gas consumption. For example, materials like silicon, aluminum, and steel used in solar panels and wind turbines have lower environmental footprints than fossil fuels like natural gas. Furthermore, adopting electricity from renewable sources promotes sustainability and reduces dependence on finite fossil resources. Before materials are altered (Scenario 1), gas consumption results in significantly lower pollution levels—24, 27, and 31% less—in indicators of human health water consumption, terrestrial ecosystem water consumption, and aquatic ecosystem water consumption, respectively, compared to electricity consumption. Hydraulic fracturing, a technique used in natural gas extraction, involves injecting substantial volumes of water, chemicals, sand, and gravel into shale formations. This process is water-intensive and can contaminate groundwater, adversely affecting human health and terrestrial and aquatic ecosystems. Additionally, the production and processing of natural gas can introduce pollutants into water sources. Inadequate wastewater management and leaks from the gas industry exacerbate water pollution, impacting human health and aquatic ecosystems. Gas consumption contributes to aquatic ecosystem degradation through pollution, habitat destruction, and alterations in water quality. Discharging wastewater from gas production and leaks can harm aquatic organisms, disrupt the aquatic food chain, and degrade water quality in rivers, lakes, and other aquatic environments²⁴. Figure 8 demonstrates that implementing material

changes reduces pollution levels from electricity and gas consumption in each indicator by approximately 6 to 1%.

Weighting

According to process weighting and Fig. 9, the human health indicator shows notable degradation. Gas consumption initially causes 9.37 kpt of pollution, which decreases to 8.95 kpt after material adjustments (Scenario 2). Meanwhile, electricity consumption begins at 5.88 kpt and decreases to 5.6 kpt after similar changes. In Fig. 9, the vertical axis shows the amount of destruction, and the horizontal axis shows the indicators. Gas emissions include pollutants that can pose more significant risks to human health than those emitted by electricity. For example, the combustion of natural gas releases nitrogen oxides, volatile organic compounds, and particulate matter, adversely affecting respiratory health and overall well-being. Gas consumption processes inherently pose more significant pollution risks to human health than electricity consumption. Factors such as emissions per unit of energy produced, types of pollutants emitted, and the proximity of gas facilities to human populations are significant contributors to these risks. The extraction and production of natural gas introduce specific health hazards for workers and local communities. These include exposure to hazardous chemicals and air and water pollution. Additionally, there is a heightened risk of accidents during drilling and fracturing operations. Overall, these hazards have a more pronounced negative impact on human health than those associated with electricity generation. Furthermore, gas extraction and processing often require chemical additives, which can pose health risks if released into the environment or encountered during production²⁵.

After implementing material changes (Scenario 2), gas emissions in the ecosystem indicator decreased from 0.82 kpt to 0.78 kpt, while electricity pollution decreased from 0.49 to 0.46 kpt. Gas combustion releases pollutants that can significantly impact ecosystems more than those emitted from electricity generation. For instance, natural gas combustion generates nitrogen oxides, sulfur dioxide, and volatile organic compounds, contributing to air pollution and potentially harming vegetation, aquatic ecosystems, and wildlife. The extraction and processing of natural gas present substantial risks of water pollution, primarily through the release of chemical additives and contamination of water sources during drilling and fracturing operations. This pollution profoundly affects aquatic ecosystems by disrupting marine life, polluting drinking water sources, and degrading water quality. Additionally, gas extraction operations require extensive infrastructure, such as well pads, pipelines, and access roads, leading to habitat destruction and fragmentation. These activities can reduce biodiversity, disturb wildlife corridors, and alter landscapes. Gas consumption has a more detrimental impact on ecosystem indicators than electricity generation due to factors like emission composition, pollution intensity, methane emissions, water pollution, and habitat destruction. These factors collectively contribute to a more significant overall negative impact on ecosystems associated with gas consumption processes than electricity generation²⁶.

In the resource indicator, gas consumption declines from 0.38 to 0.36 kpt after implementing material changes. In contrast, electricity consumption exhibits minimal pollution, dropping from 0.008 to 0.007 kpt before (Scenario 1) and after the modifications (Scenario 2). The significantly higher volume of gas consumed than electricity underscores its broader utilization, leading to a more significant impact on the resource indicator. Natural gas is a non-renewable energy resource sourced from the Earth's crust. Its extraction and consumption can diminish reserves over time, affecting the availability and sustainability of this energy source. On the

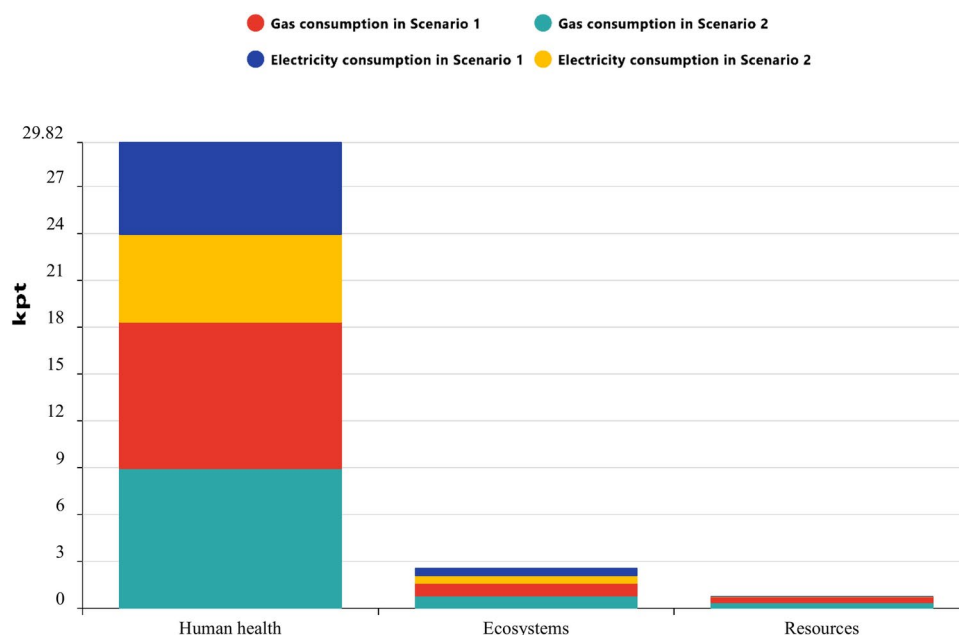


Fig. 9. Weighting of environmental impacts assessment of energy consumption.

other hand, electricity can be generated from various sources, such as solar, wind, and hydropower, with lower environmental impacts and greater long-term sustainability. Gas consumption, characterized by its substantial volume, higher pollution intensity, depletion effects, and external ecological consequences, likely has a more detrimental impact on the resource indicator than electricity. Despite electricity's lower absolute pollution levels, these factors highlight the more significant negative influence of gas consumption on the resource indicator than electricity²⁷.

Life cycle impact assessment of materials

Characterization

The findings depicted in Fig. 10 highlight a significant rise in global warming index pollution when cement mortar is used in 20-cm-thick walls, increasing sharply from 29 to 57.82% following material adjustments. Conversely, 10-cm blocks exhibit a 4.3% impact on global warming indices, which escalates to 8.7% after material changes (Scenario 2). These blocks also demonstrate the most significant effect on the human carcinogenic toxicity index, increasing from 11.6 to 22.11% post-material modifications. Overall, comparisons across all processes indicate a doubling of the environmental effects due to material changes. Additionally, incorporating insulation materials after adjustments leads to minimal adverse effects, affecting less than 1% of most indices, except for the fossil resource depletion index, which shows the highest impact at 1.7% pollution. Manufacturing cement mortar and concrete blocks result in significant CO₂ emissions, intensifying global warming by retaining heat in the Earth's atmosphere. These processes are also energy-intensive, releasing diverse air pollutants like particulate matter and nitrogen oxides, which pose risks to respiratory health. Workers in this industry face exposure to hazardous materials during production and transport, increasing health hazards. Moreover, extracting raw materials for concrete production leads to habitat destruction, soil erosion, and disturbance of local ecosystems. Land utilization linked to mining and raw material extraction exacerbates biodiversity depletion and ecological instability in terrestrial environments. Concrete production introduces pollutants such as sediments, chemicals, and contaminants into freshwater bodies, posing threats to aquatic life, ecosystem stability, and water quality. While concrete manufacturing itself does not directly impact the ozone layer, specific chemical processes involved in cement production can release ozone-depleting substances such as chlorofluorocarbons (CFCs) or hydrochlorofluorocarbons (HCFCs)²⁸.

Specific raw materials used in concrete production, such as specific aggregates and minerals, may contain naturally occurring radioactive elements that emit ionizing radiation. Moreover, releasing volatile organic compounds and nitrogen oxides during concrete production can lead to the formation of ground-level ozone, potentially harming human respiratory health. Processes involved in concrete production, such as extraction, crushing, and milling of raw materials, release fine particles that contribute to air pollution, posing risks to both human health and the environment. Additionally, burning fossil fuels for energy in concrete production emits sulfur dioxide and nitrogen oxides, which contribute to acid deposition, causing soil acidification and ecological harm in terrestrial ecosystems. Runoff and nutrient-rich wastewater originating from concrete production sites are crucial in inducing eutrophication in freshwater and marine ecosystems. These surplus nutrients promote algae growth, leading to oxygen depletion and disturbances in aquatic environments. Moreover, including chemicals and additives in concrete manufacturing poses risks to terrestrial ecosystems and organisms, influencing food chains and ecological dynamics both directly and indirectly. Drinking pollutants from concrete production into freshwater systems further endangers aquatic organisms, imperiling their survival and reproductive success²⁹.

Concrete production contributes to pollution by releasing chemicals and pollutants into marine environments, harming marine organisms and ecosystems. Workers and nearby communities may face carcinogenic risks from

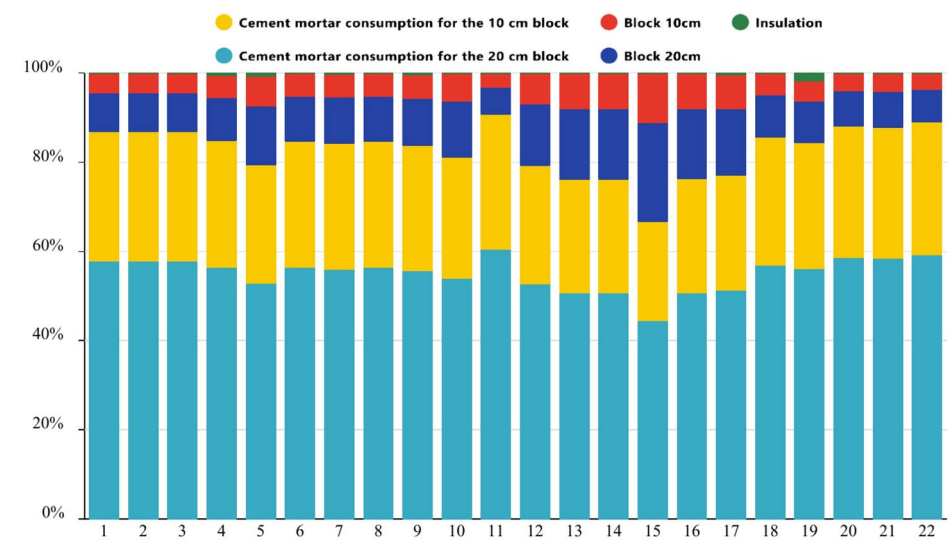


Fig. 10. Environmental impacts assessment of materials consumption based on various indicators.

exposure to certain materials used or released during production. Additionally, non-carcinogenic toxins emitted in concrete production pose health risks through inhalation or ingestion. Extracting raw materials for concrete and related mining activities alters land use, impacting habitats and natural landscapes. This includes extracting materials like sand, gravel, and limestone, which depletes mineral resources over time. Concrete production is heavily dependent on fossil fuels for electricity and heating, contributing to the depletion of these resources. Moreover, the process requires substantial water consumption for mixing, curing, and other stages, placing strain on local water sources and potentially causing scarcity that directly affects human health. Excessive water usage in concrete production also risks terrestrial ecosystems by reducing water availability for natural habitats and vegetation, leading to habitat destruction and wildlife impacts. Additionally, water extraction for concrete processes can lower river or stream flow rates and alter habitats critical for fish, amphibians, and other aquatic organisms, thus affecting aquatic ecosystems³⁰.

Weighting

Figure 11 illustrates that construction materials have significant environmental impacts, particularly concerning human health. In Fig. 11, the vertical axis shows the amount of destruction, and the horizontal axis shows the indicators. Mortar use in a 10 cm wall initially causes 4.76 kpt of pollution, increasing to 9.47 kpt after material adjustments (Scenario 2). Similarly, 10 cm blocks initially contribute 1.34 kpt of damage, which rises to 2.69 kpt post modifications. Concrete production releases various air pollutants such as particulate matter, nitrogen oxides, sulfur dioxide, and volatile organic compounds, all of which adversely affect human respiratory health and can lead to respiratory diseases. Discharge and runoff from concrete production sites may contain pollutants and chemicals that pose health risks if they enter drinking or recreational waters. The energy-intensive processes involved in concrete production, including raw material extraction, transportation, cement manufacturing, and curing, contribute significantly to environmental pollution and increased greenhouse gas emissions. Combusting fossil fuels in energy production releases pollutants that impact air quality and human health. Furthermore, extracting raw materials for concrete production can destroy habitat, soil erosion, and disrupt ecosystems³¹.

The Ecosystem Index shows that all processes produce less than 1 Kpt of adverse effects. Initially, cement mortar and 10 cm blocks cause 0.45 kpt and 0.09 kpt of pollution, respectively, which doubles following material adjustments. In the Resources Index, all processes also yield impacts of under 0.01 kpt. Insulation layers contribute less than 0.01 kpt of pollution across all indices. The extraction of raw materials for concrete production, such as sand, gravel, and limestone, typically involves mining activities. These activities can lead to habitat destruction and fragmentation, disrupting ecosystems, destroying wildlife habitats, displacing native species, and contributing to biodiversity loss and ecological imbalance. Extracting raw materials and developing concrete infrastructure can accelerate soil erosion and degrade soil quality. Erosion diminishes soil fertility, compromises water quality, and destabilizes ecosystems. This process results in sediment buildup in waterways, nutrient runoff, and the loss of cultivable land, impacting agricultural yields and natural habitats. Concrete manufacturing necessitates substantial water usage for blending, curing, and cleansing operations. Extracting water for these processes can diminish local water reservoirs, particularly in water-scarce regions or areas with competing water demands. Over-extraction can disrupt aquatic ecosystems, diminish water access for other purposes, and worsen water stress in vulnerable locales. Additionally, the energy-intensive aspects of concrete production contribute to resource scarcity and environmental deterioration. The extraction, processing, and transportation of raw materials and the production and curing processes require significant energy inputs, primarily sourced from fossil fuels. This substantial energy demand contributes to greenhouse gas emissions,

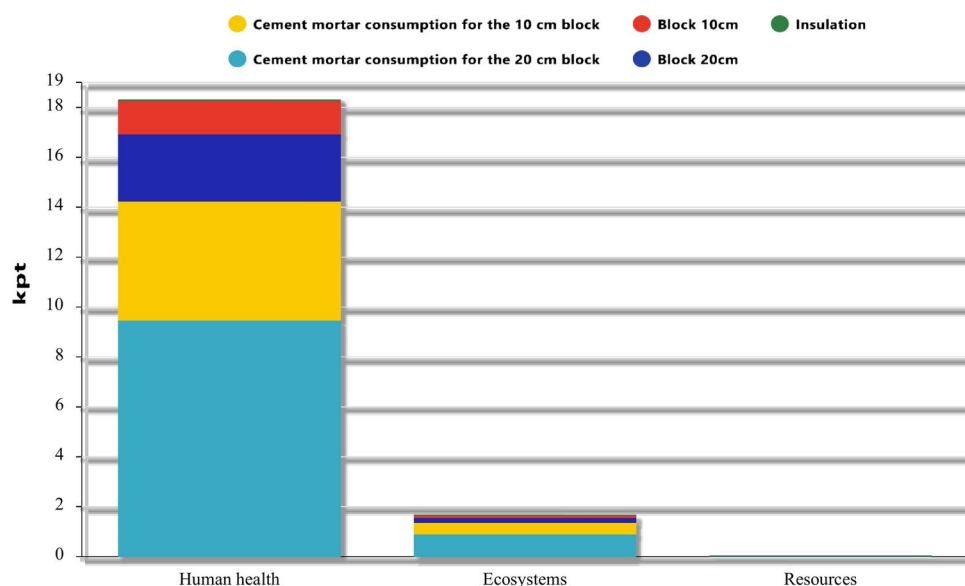


Fig. 11. Weighting of environmental impacts assessment of materials consumption.

air pollution, and climate change, profoundly affecting ecosystems and natural resources. Concrete production generates surplus materials such as unused concrete, slurry, and packaging, posing environmental risks. Improper concrete waste disposal can lead to soil and water pollution, impacting ecosystems and wildlife. Effective concrete waste management is critical for reducing its environmental footprint on resources and ecosystems³².

Comparison of life cycle assessment of energy and materials in scenario 1 and scenario

Characterization

Upon reviewing the environmental impacts depicted in Fig. 8, it becomes clear that energy consumption decreases over the building's 50-year lifespan following adjustments to materials. Concrete, concrete blocks and insulation were strategically integrated based on project-specific data detailed in Fig. 10's life cycle assessment for each material. Figure 12 comprehensively evaluates the project before (Scenario 1) and after material modifications (Scenario 2), highlighting how increased wall thickness affects both environmental impacts and the project's life cycle. In Fig. 12, the vertical axis shows the percentage of destruction, and the horizontal axis shows the indicators. Graphically depicted in Fig. 12, the project achieved energy savings despite increased material usage, resulting in a 4.5% reduction in environmental impacts across all assessed criteria. Increasing the thickness of concrete walls decreases the building's energy demand for heating and cooling, thereby reducing greenhouse gas emissions from electricity and gas consumption, which helps mitigate global warming. This reduced energy usage also lowers emissions of air pollutants and harmful substances throughout the project's duration, enhancing air quality and reducing adverse health effects. Moreover, the decline in greenhouse gas emissions aids in diminishing both air and water pollution, thereby significantly lowering impacts on soil, vegetation, and wildlife. Additionally, decreased energy consumption leads to less water use and pollutant release, which helps mitigate implications for freshwater ecosystems and contributes to protecting the ozone layer^{33,34}.

Reduced energy consumption has multiple environmental benefits. It decreases ozone precursor emissions, improving air quality and reducing health risks linked to ground-level ozone exposure. Additionally, lower energy use reduces emissions of acidic substances, mitigating soil acidification and benefiting vegetation cover. Moreover, decreased energy consumption lowers emissions of toxic substances, reducing impacts on terrestrial, freshwater, and marine ecosystems³³. Furthermore, it leads to fewer emissions of both carcinogenic and non-carcinogenic substances, thereby decreasing health risks associated with exposure to these pollutants. Increasing the thickness of concrete walls can potentially increase material usage; however, it simultaneously reduces overall energy consumption and resource extraction. This reduction in energy consumption helps decrease land use and alleviates pressure on mineral and fossil resources. Moreover, lower energy use reduces water consumption for electricity generation and minimizes pollutant discharge, thereby mitigating impacts on human health, aquatic life, and terrestrial ecosystems. The analysis indicates that implementing thicker concrete walls in construction projects yields significant energy savings and environmental benefits. These benefits include lower greenhouse gas emissions, enhanced air quality, and reduced adverse health effects. Furthermore, decreased energy consumption reduces pollution and mitigates impacts on terrestrial and freshwater ecosystems. Overall, these

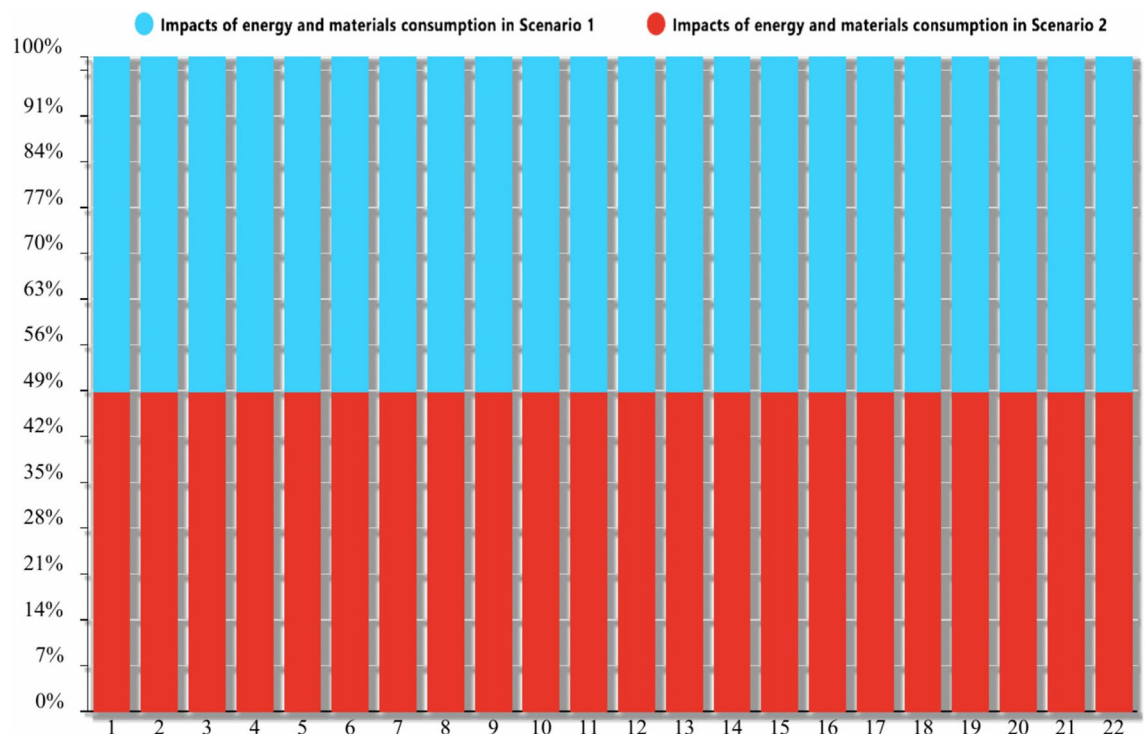


Fig. 12. Comparison of environmental impacts of energy and materials in scenario 1 and scenario 2 based on various indicators.

measures positively impact human health and the health of aquatic and terrestrial ecosystems. Increasing wall thickness and integrating insulation represent effective strategies for achieving long-term energy efficiency and reducing environmental impacts^{35,36}.

Environmental Impact assessment

In the first scenario, the walls are designed with 10 cm thick blocks of the same size. This simple and economical design may reduce construction costs in the short term, but in the long term, it will negatively impact energy consumption and indoor environmental quality. Thinner walls generally have less insulation, which leads to increased heat loss in cold seasons and an increased need for heating systems. Similarly, in hot seasons, these walls cannot prevent heat from entering the building from outside, increasing the need for cooling systems. In contrast, the second scenario offers significantly better performance by increasing the thickness of the walls to 20 cm and using insulated blocks with a layer of bitumen insulation. These changes reduce heat loss and improve the energy efficiency of the building. By reducing the need for heating and cooling systems, the total energy consumption of the building is reduced, which in turn leads to a reduction in greenhouse gas emissions. The life cycle assessment results show that the second scenario performs 5% better than the first regarding human health, ecosystem, and resource indicators. Thicker walls and proper insulation reduce the need for energy for heating and cooling. This reduces operating costs and puts less pressure on energy resources. By reducing temperature and humidity fluctuations inside the building, the quality of the living environment for residents is improved. This can have a positive impact on the mental and physical health of individuals. By reducing energy consumption, greenhouse gas emissions are also reduced. This helps preserve ecosystems and mitigate climate change. The second scenario is more economically and environmentally sustainable by choosing materials with longer lifespans and higher performance. Overall, this analysis shows that investing in building designs with better energy performance and insulation not only leads to cost savings over the life of the building but also has significant positive impacts on human health and the environment. Choosing the right materials and practical design can play a prominent role in creating high-performance and environmentally friendly buildings. Figure 13 shows the environmental impact assessment of both scenarios^{37,38}. In Fig. 13, the vertical axis shows the percentage of destruction, and the horizontal axis shows the indicators.

Normalization

Normalizing the two cleanroom building wall design scenarios shows significant impacts on human health, ecosystem, and resource indices. Figure 14 clearly shows these differences in the environmental performance of the two scenarios. In Fig. 14, the vertical axis shows the amount of destruction, and the horizontal axis shows the indicators. In the first scenario, with 10 cm thick, thin walls and blocks of the same size, the building performance in the three indicators is affected. The first scenario score is 38,169.4648 for the human health index, 3285.3653 kpt for the ecosystem and 1940.9853 kpt for the resources. These numbers indicate high energy consumption and negative impacts on the health of the occupants and the environment. Impact on human health: Thin walls and lack of proper insulation lead to temperature fluctuations inside the building, leading to respiratory problems and other diseases. These temperature fluctuations can also affect the quality of sleep and comfort of the occupants. Impact on the ecosystem: As the need for heating and cooling systems increases, energy consumption in the first scenario rises, which leads to higher greenhouse gas emissions and tremendous pressure on ecosystems. A lower score on this index (3285.3653 kpt) indicates negative impacts on biodiversity and climate change. Impact on resources: Scenario 1, with a score of 1940.9853 kpt on the resource index, indicates a high consumption of natural resources to supply the building's energy needs. This leads to a decrease in natural resources and increases operating costs. In contrast, the second scenario offers better performance

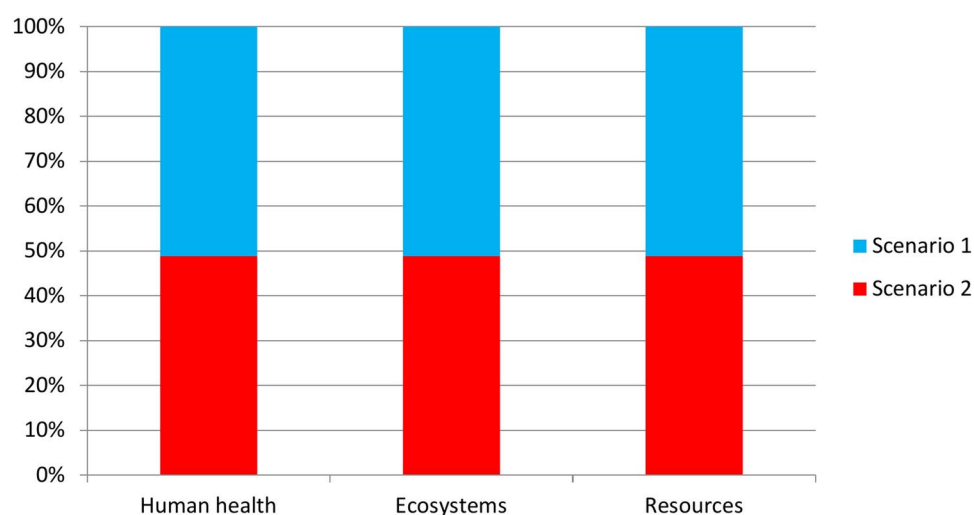


Fig. 13. Specialization of environmental impacts assessment of energy and materials in scenario 1 and scenario 2.

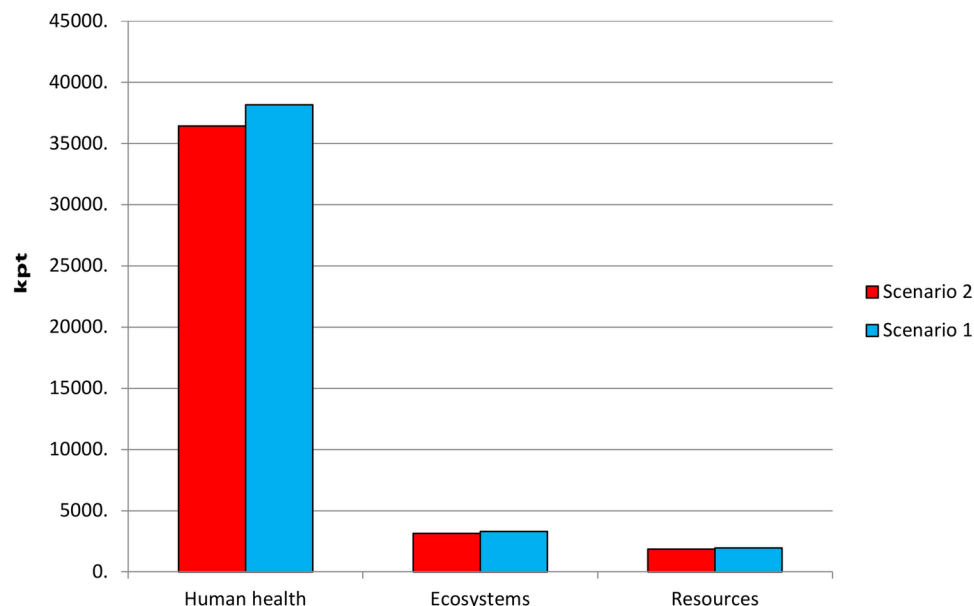


Fig. 14. Normalization of environmental impacts assessment of energy and materials in scenario 1 and scenario 2.

with thicker walls of 20 cm insulated blocks and a bitumen insulation layer. The score for this scenario is 36,435.8516 kpt for the human health index, 3136.3816 kpt for the ecosystem, and 1854.8016 kpt for the resources³⁹. These numbers indicate reduced energy consumption and positive effects on human health and the environment. Impact on human health: Thicker walls and proper insulation lead to a more stable environment in terms of temperature and humidity, improving residents' quality of life. The likelihood of diseases is reduced by reducing temperature fluctuations, and the residents' comfort is increased. Impact on the ecosystem: By reducing the need for heating and cooling systems in the second scenario, energy consumption is reduced, which leads to a reduction in greenhouse gas emissions. A higher score on this index (3136.3816 kpt) indicates a positive impact on biodiversity and a reduction in adverse effects on ecosystems. Impact on Resources: The second scenario, with a score of 1854.8016 kpt on the Resource Index, represents a more efficient use of natural resources. Reducing energy requirements reduces pressure on natural resources, which helps conserve resources and reduces operating costs. Overall, the comparison of the two scenarios shows that effective designs using insulating materials and thicker walls not only lead to cost savings over the life of the building but also have significant positive impacts on human health, the ecosystem, and resources. Investing in sustainable and energy-efficient designs can play a key role in creating high-performance and environmentally friendly buildings^{40,41}.

Weighting

Figure 15 compares the environmental weighting of energy and materials before (Scenario 1) and after the changes (Scenario 2). In Fig. 15, the vertical axis shows the amount of destruction, and the horizontal axis shows the indicators. The analysis of the project's environmental impact assessment reveals the most significant adverse effects on the human health index. Initially, the clean room facility recorded 15.26 kpt pollution in this category, which decreased to 1 kpt after adjusting materials (Scenario 2). Regarding the ecosystem, impacts are most pronounced following the human health index, with an initial degradation of 1.31 kpt reducing to 0.1 kpt post-adjustment. The resources index shows the least pollution, where the clean room facility initially contributes 0.38 kpt, decreasing to 0.01 kpt after material modifications (Scenario 2). Enhancing building insulation to reduce energy consumption decreases greenhouse gas emissions and improves air quality. This improvement can alleviate respiratory diseases, cardiovascular issues, and other health problems associated with poor air quality, positively impacting human health. Additionally, lower energy usage alleviates pressure on natural resources and reduces pollution, benefiting terrestrial and aquatic ecosystems. Decreased demand for fossil fuels and minerals further diminishes extraction and discharge impacts, contributing to the conservation of natural habitats and minimizing environmental disruptions linked to resource extraction⁴².

Human health exhibits the highest pollution levels among these three indicators, primarily because of air pollution's direct and immediate impact. Even minor increases in pollution levels can markedly affect human health, underscoring its heightened sensitivity relative to ecosystems and resources. Consequently, enhancing air quality and curbing greenhouse gas emissions, as opposed to their effects on ecosystems and resources, promptly yields substantial positive benefits for human health.

Final score

In the life cycle analysis of energy and materials consumption for two scenarios, the results obtained from Fig. 16 indicate significant differences in the scores of the three leading indicators: human health, ecosystems, and

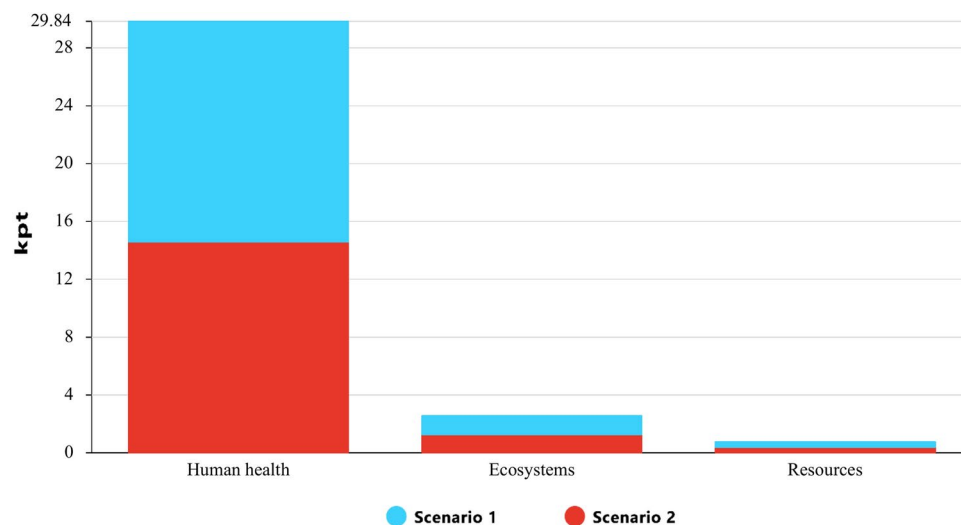


Fig. 15. Weight comparison of the environmental impacts of energy and materials in scenario 1 and scenario 2.

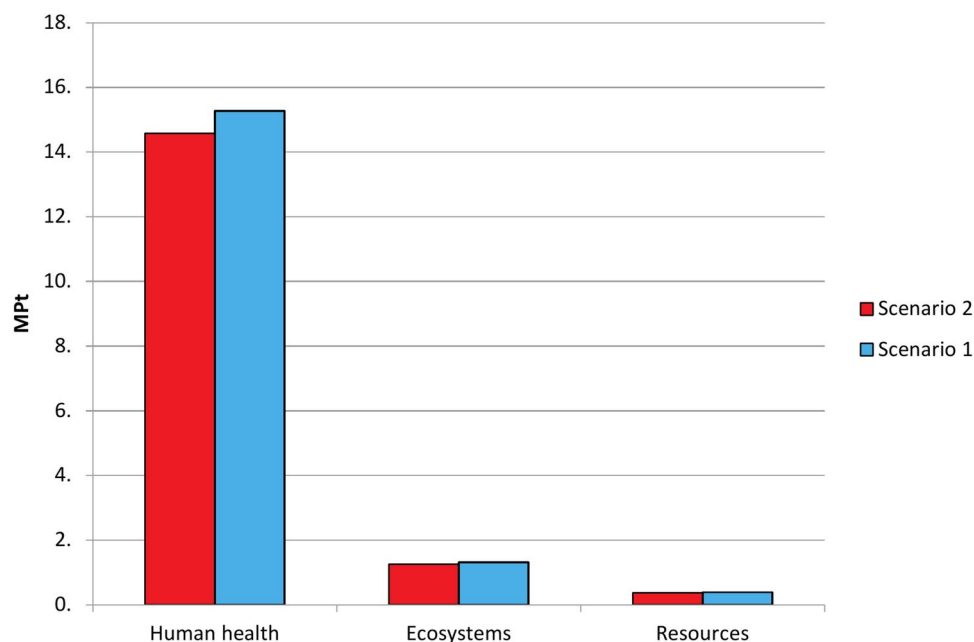


Fig. 16. Final score of environmental impacts assessment.

resources. In Fig. 16, the vertical axis shows the score value, and the horizontal axis shows the indicators. The first scenario's score for the human health index is 15.2678 Mpt. This higher score indicates the design's harmful effects on the occupants' health. Thinner walls 10 cm thick and using similar blocks cause more significant temperature fluctuations inside the building, leading to respiratory problems and reduced indoor air quality⁴¹. In contrast, the second scenario, with a score of 14.5743 Mpt, indicates improved human health. Thicker walls 20 cm thick and a bitumen insulation layer reduce temperature fluctuations and improve indoor air quality, which helps to increase the comfort and health of the occupants. The first scenario has a score of 1.3141 Mpt for the ecosystems index. This score indicates the negative impacts of the design on the environment, as the high energy demand for heating and cooling leads to increased greenhouse gas emissions. In contrast, the second scenario, with a score of 1.2546 Mpt, indicates a reduction in the negative impacts on ecosystems. With more efficient design and a reduced need for heating and cooling systems, energy consumption is reduced, which helps reduce harmful gas emissions. The first scenario has a score of 0.3882 Mpt in the resources index, which indicates a high consumption of natural resources. Inefficient building design leads to increased energy consumption and pressure on natural resources⁴³. On the other hand, the second scenario, with a score of 0.371 Mpt, indicates an optimization of the consumption of natural resources. Using high-quality materials

and proper insulation reduces energy requirements and thus puts less strain on natural resources. Overall, the comparison of the two scenarios shows that effective design using insulating materials and thicker walls not only leads to cost savings over the life of the building but also has significant positive impacts on human health, ecosystems, and resources. The second scenario, with its improved features, not only increases residents' quality of life but also helps preserve the environment. These results show that the choice of appropriate materials and design can directly impact the quality of life of residents and the preservation of the environment. Investing in sustainable and energy-efficient designs can play a key role in creating high-performance buildings. Therefore, paying attention to the details of the design and choosing high-quality materials is important not only from an economic perspective but also from a social and environmental perspective⁴⁴.

Sensitivity analysis

The Monte Carlo diagram in Figs. 17a,b illustrates uncertainty evaluation across various indicators associated with the relevant project processes. The findings show that scenario two's uncertainty level is significantly reduced. Notably, indicators 20, 21, and 22 identify the highest levels of uncertainty, which pertain to human drinking water, terrestrial ecosystems, and aquatic ecosystems. Assessing water consumption from drinking water ecosystems is crucial for human health, as it involves analyzing potential impacts on water resources that directly influence the quality of water available for consumption. Uncertainty in this indicator may stem from various factors, including the quality of the water source, the treatment methods utilized to ensure its safety and the likelihood of pollutants being introduced during construction activities. The indicator for water consumption in terrestrial ecosystems emphasizes evaluating water usage by surrounding ecosystems, including vegetation, throughout the construction phase. The uncertainty associated with this indicator can fluctuate based on several elements such as the type of ecosystem, climatic conditions, specific project characteristics like the location of construction, types of vegetation present, and the availability of water resources. Additionally, alterations in weather patterns, seasonal changes, and the effectiveness of water management strategies can further contribute to uncertainties tied to this indicator⁴⁵. In contrast to the first two indicators, the water consumption of aquatic ecosystems typically demonstrates a greater degree of uncertainty. This heightened uncertainty can be attributed to the intricate nature of aquatic ecosystems, the variety of species reliant on these environments, and the potential environmental repercussions of construction activities. Aquatic ecosystems tend to be more susceptible to fluctuations in both water quality and quantity, complicating efforts to forecast potential impacts and levels of water consumption accurately. Moreover, these ecosystems are frequently interconnected with other ecological systems, which can amplify uncertainties⁴⁶. Other indicators also reveal uncertainties, including ionizing radiation, freshwater ecotoxicity, marine ecotoxicity, carcinogenic toxicity, and non-carcinogenic toxicity in humans. The uncertainty associated with the freshwater ecotoxicity indicator can be influenced by chemical pollutants released during construction, dilution capacity, water flow rates, and local ecological conditions. Marine ecotoxicity considers the potential adverse effects of construction activities on aquatic ecosystems; uncertainties in this area arise from various factors such as pollutant releases, affected marine species and habitats, possible bioaccumulation of pollutants, and the efficacy of mitigation measures designed to minimize impacts. Uncertainties related to carcinogenic and non-carcinogenic toxicity indicators in humans can be linked to factors such as the types and concentrations of harmful substances released, exposure duration and frequency, individual susceptibility to toxins, and the effectiveness of existing protective measures. The highest levels of uncertainty are noted in indicators concerning human drinking water, terrestrial ecosystems, and aquatic ecosystems. These uncertainties originate from factors including water source quality, treatment processes, potential pollutant introduction, ecosystem type, climatic conditions, project-specific characteristics, changes in weather patterns, and the effectiveness of water management practices. The intricate nature and sensitivity of aquatic ecosystems further exacerbate the uncertainties associated with levels of water consumption⁴⁷.

Conclusion

Assessing the environmental impacts of energy consumption in the clean room construction project, mainly through increased wall thickness to reduce energy usage, offers crucial insights into the sustainability implications of construction practices. Understanding these impacts is vital for addressing environmental and human health risks linked to building projects. Initial assessments revealed substantial pollution levels. Greenhouse gas emissions accounted for 33% and 18.06% of total gas and electricity consumption, respectively. Terrestrial ecosystems and freshwater ecosystems experienced degradation rates of 13.5% and 5%, respectively, in environmental toxicity indices. Moreover, indices measuring marine ecosystem toxicity, human carcinogenic toxicity, and human non-carcinogenic toxicity related to electricity consumption were 10% lower than those associated with gas consumption before material adjustments (Scenario 1). Gas consumption entails burning natural gas, releasing pollutants such as nitrogen oxides, sulfur dioxide, and particulate matter into the atmosphere. Electricity consumption exhibits lower environmental degradation in mineral resources and fossil fuel scarcity indices, contributing only 5% and 1% pollution, respectively. Before material adjustments (Scenario 1), gas consumption led to pollution levels that were 24%, 27%, and 31% higher in indices measuring human health water consumption, terrestrial ecosystem water consumption, and aquatic ecosystem water consumption, respectively, compared to electricity consumption. After implementing material (Scenario 2) changes, pollution emissions from electricity and gas consumption in each index decreased by approximately 6 to 1%.

The life cycle assessment of materials before (Scenario 1) and after modifications (Scenario 2) reveals significant findings. Using 20 cm thick walls leads to the highest increase in the global warming index, with degradation rising from 29% to 57.82% due to cement mortar consumption during construction. Similarly, 10 cm blocks show a rise in degradation from 4.3% to 8.7% in the global warming index following material adjustments. Furthermore, these blocks exhibit the most substantial impact on the human carcinogenic toxicity

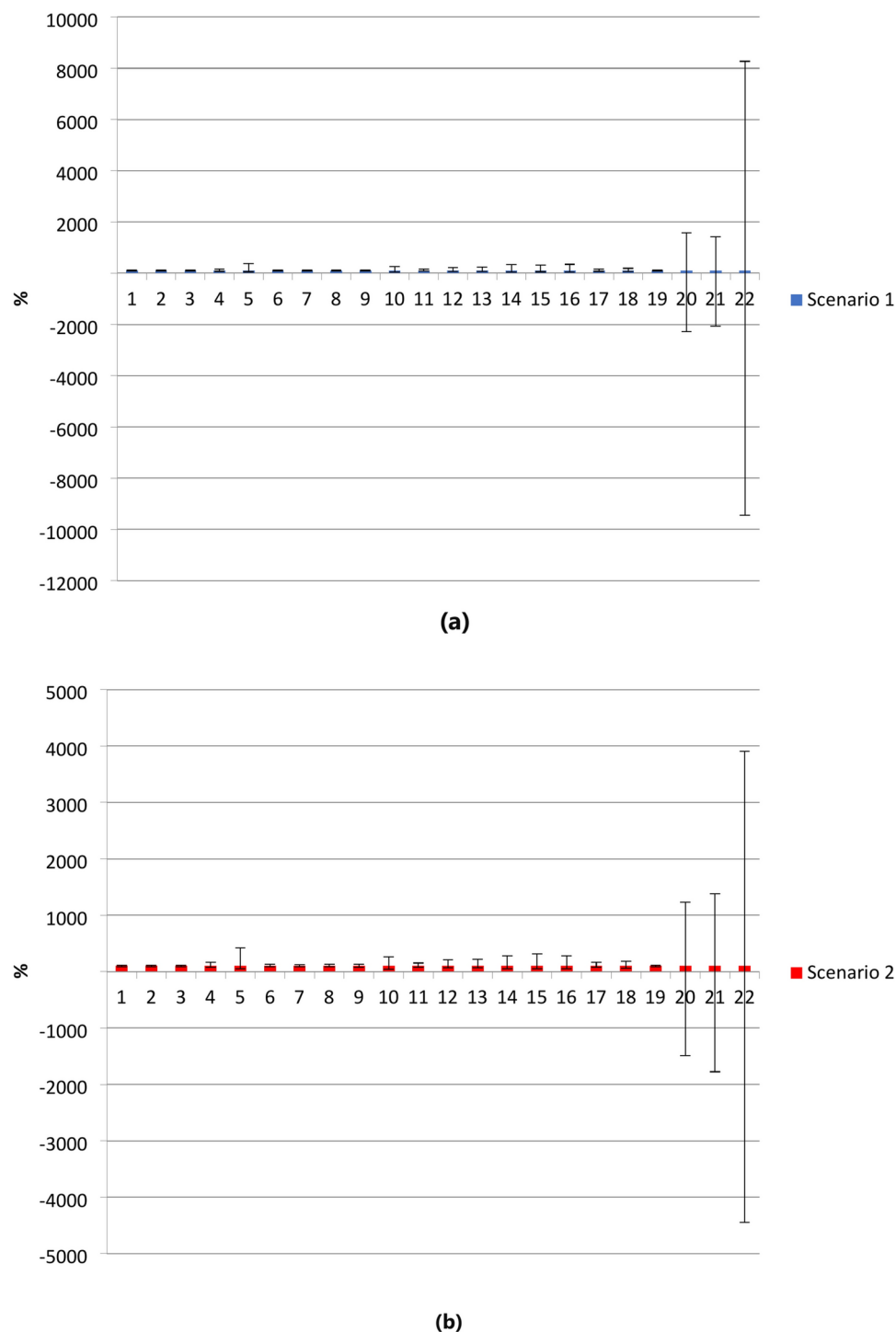


Fig. 17. (a) Uncertainty of environmental impacts using the Monte Carlo diagram in scenario 1; (b) Uncertainty of environmental impacts using the Monte Carlo diagram in scenario 2.

index, increasing from 11.6% to 22.11% after material changes. Overall comparisons across all processes indicate doubling the environmental effects associated with material consumption after modifications. Additionally, concrete and cement production releases significant amounts of CO₂ and other pollutants, adversely affecting human respiratory health, aquatic and terrestrial life, marine ecosystems, the ozone layer, and biodiversity.

The life cycle assessment of the case study project before (Scenario 1) and after material changes (Scenario 2) has evaluated both energy consumption and environmental impacts. Analysis indicates that increasing the thickness of concrete walls and adding insulation layers in a clean room construction project can substantially lower energy use and environmental effects throughout the building's 50-year lifespan. Despite the higher material usage, there has been an overall positive effect on energy consumption, leading to a 4.5% reduction

in environmental impacts across all metrics. By decreasing energy consumption, benefits extend beyond reducing greenhouse gas emissions to improving air quality and minimizing health risks associated with pollution in terrestrial and aquatic ecosystems. Implementing such strategies supports conserving natural resources, ecosystems, and human health. Therefore, increasing wall thickness and insulation is a practical long-term approach to enhancing energy efficiency and mitigating environmental impacts. This study emphasizes the essential integration of long-term sustainability considerations in construction projects. It highlights the potential advantages of bolstering wall thickness and employing insulation to address environmental concerns over a building's 50-year lifespan. In addition to changing the wall thickness, adding a layer of quality insulation significantly reduced the thermal and cooling load in the cleanroom construction project. This measure led to a significant reduction in energy consumption and improved the thermal efficiency of the building. Thus, greenhouse gas emissions were reduced by reducing the need for heating and cooling systems.

The results of the life cycle analysis of energy and materials consumption for the two scenarios show that the second scenario, with a score of 14.5743 Mpt in the human health index, is 0.6935 Mpt lower than the first scenario (15.2678 Mpt) and offers a significant improvement in indoor air quality and reduced temperature fluctuations. Also, in the ecosystems index, the second scenario, with a score of 1.2546 Mpt compared to the first scenario's 1.3141 Mpt, shows a reduction of 0.0595 Mpt in negative environmental impacts. Finally, in the resources index, the second scenario, with a score of 0.371 Mpt, shows an improvement of 0.0172 Mpt compared to the first scenario (0.3882 Mpt). This data shows that effective design and the use of high-quality materials lead to cost savings but also help improve human health, reduce environmental impacts, and optimize the use of natural resources.

This project's life cycle assessment results, located in Tehran, can serve as a model for Mediterranean and ocean climates. Given the similarities in precipitation and temperature patterns in these climates, mainly characterized by hot summers and mild winters, the findings from this study conducted in Tehran may apply to other regions with similar climatic conditions. Researchers can deepen their knowledge of sustainable construction methods by focusing on efficient energy practices and integrating life cycle assessments. It is crucial to prioritize support for energy-efficient initiatives and sustainable practices that improve building durability and longevity. Emphasizing the link between building lifespan and energy efficiency highlights the environmental benefits of reducing energy consumption in construction. Clear and precise communication of environmental impact data from materials and energy use is essential. Disseminating this information effectively empowers stakeholders—researchers, policymakers, and practitioners—to make informed decisions that minimize environmental footprints and advance sustainable development. Adhering to these principles and emphasizing long-term sustainability offers valuable insights into sustainable construction practices, promoting an environmentally conscious approach to building design and construction.

Data availability

Availability of data and materials The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

Pouya Partonia contributed to the conceptualization, design, data collection, and manuscript drafting. Sobhan Heidarian was involved in data analysis and critical revisions of the manuscript. Mohamadreza Sharifi assisted in data collection and provided valuable insights during the drafting process. Ghorbanali Dezvareh contributed

to the study's design and critically reviewed the manuscript. Ahmad Khodadadi Darban played a key role in data analysis and final approval of the manuscript. All authors reviewed and approved the final version, ensuring a collaborative effort throughout the research process.

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

This study was conducted according to ethical standards. The research adhered to ethical guidelines, ensuring that all participants' rights and welfare were prioritized throughout the study.

Consent to participate

Informed consent was obtained from all participants before their involvement in the study. Each participant was provided with comprehensive information regarding the study's purpose, procedures, risks, and benefits, and they voluntarily agreed to participate, ensuring their autonomy was respected.

Consent to publish

All authors have consented to the publication of this paper. Participants involved in the study were informed that their data might be published in a scientific journal, and they provided their explicit consent for this purpose.

Additional information

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